A.I.R.A. Class Print Packet

Print this file DOUBLE-SIDED, flip on the LONG edge, letter paper. Every sheet has a front and a back, so the page count is even. Pages with a blank back are meant to be blank.

Class Race (Level 3) and the Human Neural Network cards (Level 6) now run in the Field Log at badge.goss.quest. Print only sheets 1 to 3. Sheets 4 to 10 are the BACKUP paper versions, for a day when the Wi-Fi or devices fail.

Copies assume 4 squads of 5 scouts. Adjust to your roster. Scout passports are a separate file: Print_Per_Scout_Packet.pdf (landscape, flip on the SHORT edge).

Sheet	Copies	Contents
1	1	Counselor guides: Class Race / Human vs. Machine (front), Human Neural Network (back)
2	1	Counselor only: Human vs. Machine trivia (front), Timeline key (back)
3	4	Timeline cards, one set per squad. Front and back line up: print double-sided, flip on the long edge, then cut
BACKUP ONLY (print if the Wi-Fi or devices fail)		
4	4	Machine rule card and bin labels, one per squad
5	4	Card deck sheet 1, cut on dotted lines
6	4	Card deck sheet 2, cut on dotted lines
7	4	Human vs. Machine scorecard
8	4	Role badges and clue word bank
9	4	Mystery cards, cut on dotted lines
10	4	Human Neural Network scorecard

Human vs. Machine: counselor guide

8 minutes · whole class

The idea

One scout plays the Machine and follows the rules exactly. A teammate plays the Human and sorts by instinct while also answering trivia. The squad compares speed and mistakes, then talks about where machines win and where people are still needed.

The three rules (also on the Machine Rule Card)

- TRIGGER: if the card is RED.
- TASK: put it in bin A. (Anything not red goes in bin B.)
- VARIABLE: add one to the red count (a tally mark on the scorecard).

Print and prep, per squad

- 1 deck of 24 cards (pages 3 and 4). Cut on the dotted lines. Shuffle. Cards: 10 red, 5 blue, 5 green, 3 yellow, 1 pink.
- 2 bin labels, A and B (page 2). Fold in half or tape to paper plates, cups or tape marks on the floor.
- 1 Machine Rule Card (page 2) and 1 scorecard (page 6), one per squad.
- 1 trivia list for you (page 5). Read one question at a time, in order.
- A phone timer. No other supplies.

How to run it (RELAY: squads line up behind their bins)

- Round 1 (about 2 min): Scout 1 is the MACHINE. Scout 2 is the HUMAN. The Machine sorts the whole deck following the rules, exactly, while the squad times it. Then the Human sorts a reshuffled deck while you read trivia questions out loud. The Human must answer each one out loud before taking the next card, and is told only: "Put the red cards in bin A."
- Count for each: seconds to finish, cards in the wrong bin, and the red count. The squad checks: did the red count match the real number of red cards?
- Round 2 (about 2 min): swap. Pick two new scouts. Repeat. Everyone who is not playing is the timer, counter and judge.
- The exception (about 1 min): include the PINK card. Ask: "Is pink red?" Watch what the Machine does (follows the rules exactly, so pink goes in bin B, or it freezes). What does the Human do?

Score it (suggested)

- Squads earn 5 XP for a complete scorecard and a style point for the best explanation of where the machine was better or worse. Initial Passports after the debrief.

Debrief (the last 2 minutes)

- Where was the machine better? (Consistent, tireless, accurate count.)
- Where was it worse? (Exceptions, like the pink card. No judgment.)
- Why do real workflows still need humans? (Someone writes the rules and handles surprises.)
- Tie back to the Level 3 Boss Challenge: what safeguard would catch a mistake?

Human Neural Network guide

5 minutes · squads of 5 · standing

The idea

A real AI model guesses from clues. Your squad becomes a tiny model: two scouts see a card and call out clues (the Input layer), two weigh the clues and vote (the Hidden layer), and one announces the guess (the Output layer). The model only knows what it was taught, so a misleading card shows how AI gets fooled.

Print and prep, per squad

- Page 2: cut out the 5 role badges and the clue word bank. Page 3: cut the 12 mystery cards (cards 1 and 2 are the training examples). Page 4: one scorecard.
- Shuffle nothing. The cards are already in play order, and card 12 (the red ball) is the trick card. Keep the answer key below to yourself.
- Two classes the model must learn: A = FRUIT and B = BALL.

Roles (squad stands in a line)

- INPUT 1 and INPUT 2 see the card and each call out ONE clue word, quietly, from the word bank.
- HIDDEN 1 and HIDDEN 2 do not see the card. They weigh the two clues and whisper a vote, A or B, to Output.
- OUTPUT hears both votes and announces the squad's guess: Fruit or Ball.

How to run it

- Train the model first (30 seconds): show the whole squad cards 1 and 2 face up. Card 1 is Fruit (A) and card 2 is Ball (B). These two are ALL the model has ever seen. Put them away.
- Rounds 1 to 9 (cards 3 to 11): you flip a card to the Input scouts only. They give clues, Hidden votes, Output guesses. You say right or wrong. Record it on the scorecard. Rotate roles every 3 rounds. Aim for about 20 seconds a round.
- Round 10 (card 12), the trick card: the red ball. Round, red, shiny, so the model will probably guess Fruit and be wrong. Do not warn them.
- Ask: why did the model get it wrong? What clues were missing, like bounces, no stem, no leaf?

Debrief (the last minute)

- A model only knows what its data showed it. If every training fruit was red and round, red and round means fruit.
- Connect to Level 7: when you build your Teachable Machine, vary your samples so it cannot rely on one misleading clue.
- Connect to Level 4: if the training data is unfair or too narrow, the model's guesses are too.

Answer key (card number: answer)

1: Apple (Fruit) 2: Soccer ball (Ball) 3: Banana (Fruit) 4: Basketball (Ball) 5: Lemon (Fruit) 6: Tennis ball (Ball) 7: Orange (Fruit) 8: Beach ball (Ball) 9: Cherries (Fruit) 10: Baseball (Ball) 11: Strawberry (Fruit) 12: Red ball (Ball)

Trivia for the Human

counselor only · read one at a time

The Human answers each question out loud before touching the next card. Skip any you do not like. Answers are in the right column.

1. What is the Scout motto?	**Be Prepared.**
2. How many points are in the Scout Law?	**12.**
3. What do you call the knot used to join two ropes of equal size?	**A square knot (reef knot).**
4. What planet is closest to the Sun?	**Mercury.**
5. How many sides does a hexagon have?	**6.**
6. What gas do plants take in from the air?	**Carbon dioxide.**
7. What is 7 times 8?	**56.**
8. What is the capital of Florida?	**Tallahassee.**
9. What is the largest ocean?	**The Pacific.**
10. How many minutes are in two hours?	**120.**
11. What do the letters in the word AI stand for?	**Artificial intelligence.**
12. What is the freezing point of water in Fahrenheit?	**32 degrees.**
13. Name the first two points of the Scout Law.	**Trustworthy and Loyal.**
14. How many continents are there?	**7.**
15. What is the opposite of north on a compass?	**South.**
16. What is 15 plus 27?	**42.**
17. What is the hardest natural material?	**Diamond.**
18. How many legs does a spider have?	**8.**
19. Which direction does the sun rise?	**East.**
20. How many players are on a standard soccer team on the field?	**11.**
21. What color do you get from mixing blue and yellow?	**Green.**
22. How many days are in a leap year?	**366.**
23. What do bees make?	**Honey.**
24. What is 100 divided by 4?	**25.**

AI TIMELINE: COUNSELOR KEY

Human timeline (Requirement 2e)

Correct order, oldest to newest

1.	1950	Turing's paper	Alan Turing's paper started the Turing test idea.
2.	1956	Dartmouth workshop	The workshop at Dartmouth College is seen as the birth of AI as a fiel...
3.	1966	ELIZA chatbot	Joseph Weizenbaum's ELIZA showed how easily people trust a chatbot.
4.	1997	Deep Blue wins	Deep Blue showed a machine could beat the best human at a hard game.
5.	2011	Watson on Jeopardy!	IBM's Watson showed machines could handle questions in plain English.
6.	2014	Alexa and Echo	Amazon's Echo and Alexa brought voice AI into homes.
7.	2016	AlphaGo wins	AlphaGo learned by practicing against itself, a big step for machine l...
8.	2022	ChatGPT launches	ChatGPT put powerful AI in the hands of everyday people.

How to run it (5 min, whole room)

- Print one double-sided set per squad (4 squads = 4 sets). Cut on the dashed lines. Eight cards per squad.
- Squads get a shuffled stack, deal the cards out (one or two each), and line up along a wall, oldest to newest.
- Squads may not look at the backs until they say "locked." Then everyone flips their card to check.
- Award bonus XP for a correct order (suggestion: 5 XP for correct on the first try). Suggest 1 style point for teamwork.
- Each scout still writes their own five milestones on the site or worksheet (Requirement 2e). The cards are the movement game.

Discussion

- Why was there a long gap between 1956 and 1997? (Progress was slower than hoped, funding dried up in the so-called AI winters, and computers had far less power and data.)
- What changed by 2022 so that everyday people could use AI directly? (A chat interface anyone can use, no coding needed.)
- Why cross-check a date a chatbot gives you? (It can state a wrong year with total confidence.)

Print note: double-sided, flip on the LONG edge. The back page is mirrored so fronts and backs line up.

A.I.R.A. TIMELINE — Where do I go in line?

Talking to a machine at home

A smart speaker called Echo launches with a voice assistant named Alexa. Asking a machine for the weather becomes normal.

Flip the card over only after your squad has lined up.

A.I.R.A. TIMELINE — Where do I go in line?

The field gets its name

A summer workshop of researchers at a college in New Hampshire gives the field its name: artificial intelligence.

Flip the card over only after your squad has lined up.

A.I.R.A. TIMELINE — Where do I go in line?

AI you can chat with

A chatbot called ChatGPT opens to the public. Within weeks, millions of everyday people are asking an AI to write, explain, and brainstorm.

Flip the card over only after your squad has lined up.

A.I.R.A. TIMELINE — Where do I go in line?

A machine beats the chess champion

A computer beats the reigning world chess champion in a match. The champion is Garry Kasparov. The computer is IBM's Deep Blue.

Flip the card over only after your squad has lined up.

A.I.R.A. TIMELINE — Where do I go in line?

Can machines think?

A scientist asks whether machines can think. He proposes a test: a judge chats with a hidden human and a hidden machine and tries to tell which is which.

Flip the card over only after your squad has lined up.

A.I.R.A. TIMELINE — Where do I go in line?

A machine masters Go

A program called AlphaGo beats champion Lee Sedol at Go, a board game with far too many moves to check them all.

Flip the card over only after your squad has lined up.

A.I.R.A. TIMELINE — Where do I go in line?

The first chatbot

A program called ELIZA "talks" like a therapist at MIT by matching patterns in what you type. It understands nothing, but people feel heard.

Flip the card over only after your squad has lined up.

A.I.R.A. TIMELINE — Where do I go in line?

A computer wins the quiz show

A computer named Watson beats two top champions on the TV quiz show Jeopardy!, answering clues written in everyday language.

Flip the card over only after your squad has lined up.

1956

Dartmouth workshop

The workshop at Dartmouth College is seen as the birth of AI as a field.

A.I.R.A.

2014

Alexa and Echo

Amazon's Echo and Alexa brought voice AI into homes.

A.I.R.A.

1997

Deep Blue wins

Deep Blue showed a machine could beat the best human at a hard game.

A.I.R.A.

2022

ChatGPT launches

ChatGPT put powerful AI in the hands of everyday people.

A.I.R.A.

2016

AlphaGo wins

AlphaGo learned by practicing against itself, a big step for machine learning.

A.I.R.A.

1950

Turing's paper

Alan Turing's paper started the Turing test idea.

A.I.R.A.

2011

Watson on Jeopardy!

IBM's Watson showed machines could handle questions in plain English.

A.I.R.A.

1966

ELIZA chatbot

Joseph Weizenbaum's ELIZA showed how easily people trust a chatbot.

A.I.R.A.

Machine Rule Card and bin labels

one per squad · cut and fold

MACHINE RULE CARD

Follow these rules EXACTLY. Do not think. Do not guess.

TRIGGER	**If the card is RED**
TASK	**Put it in bin A. (not red: bin B)**
VARIABLE	**Add one to the red count.**

A

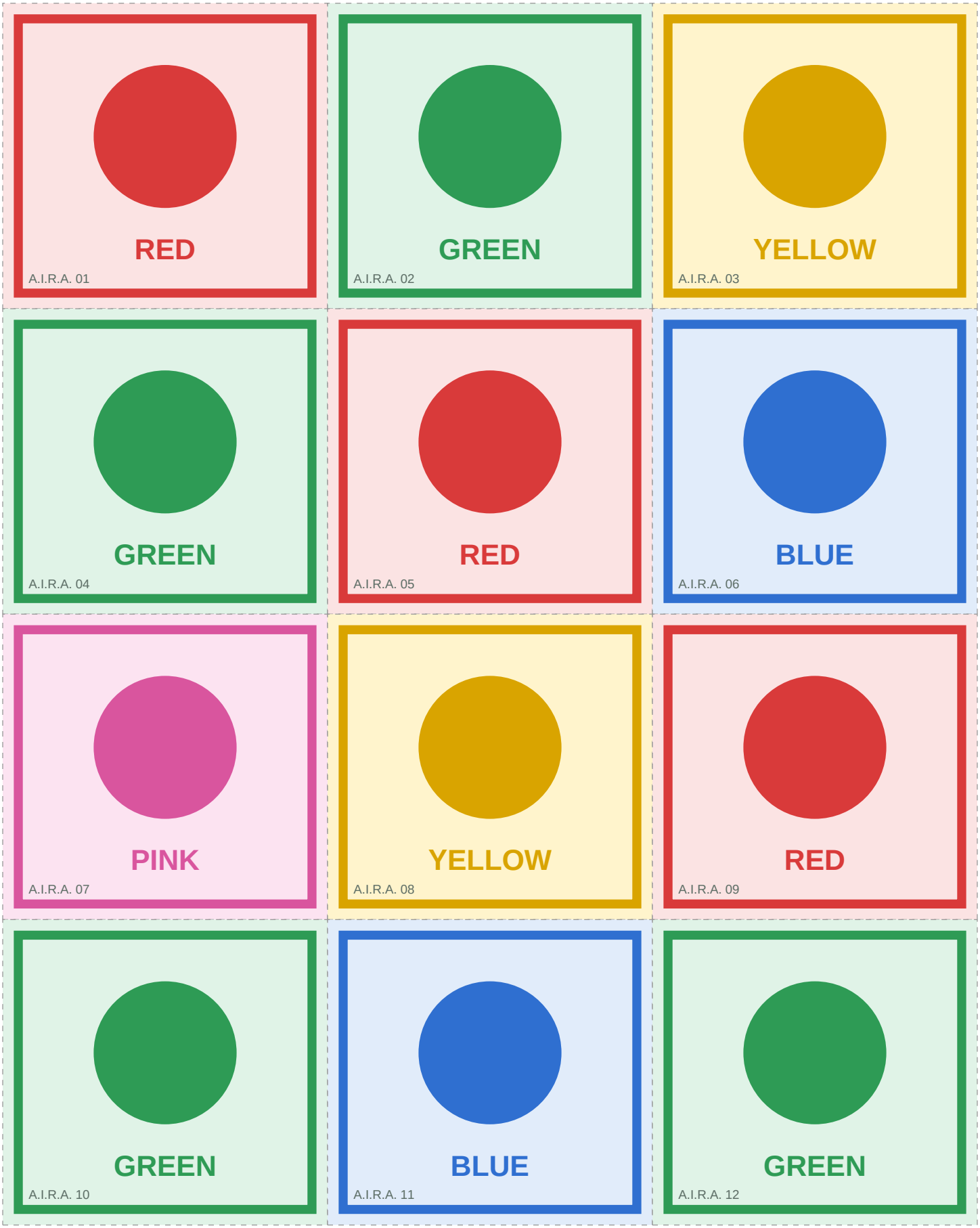
BIN A · RED CARDS GO HERE

B

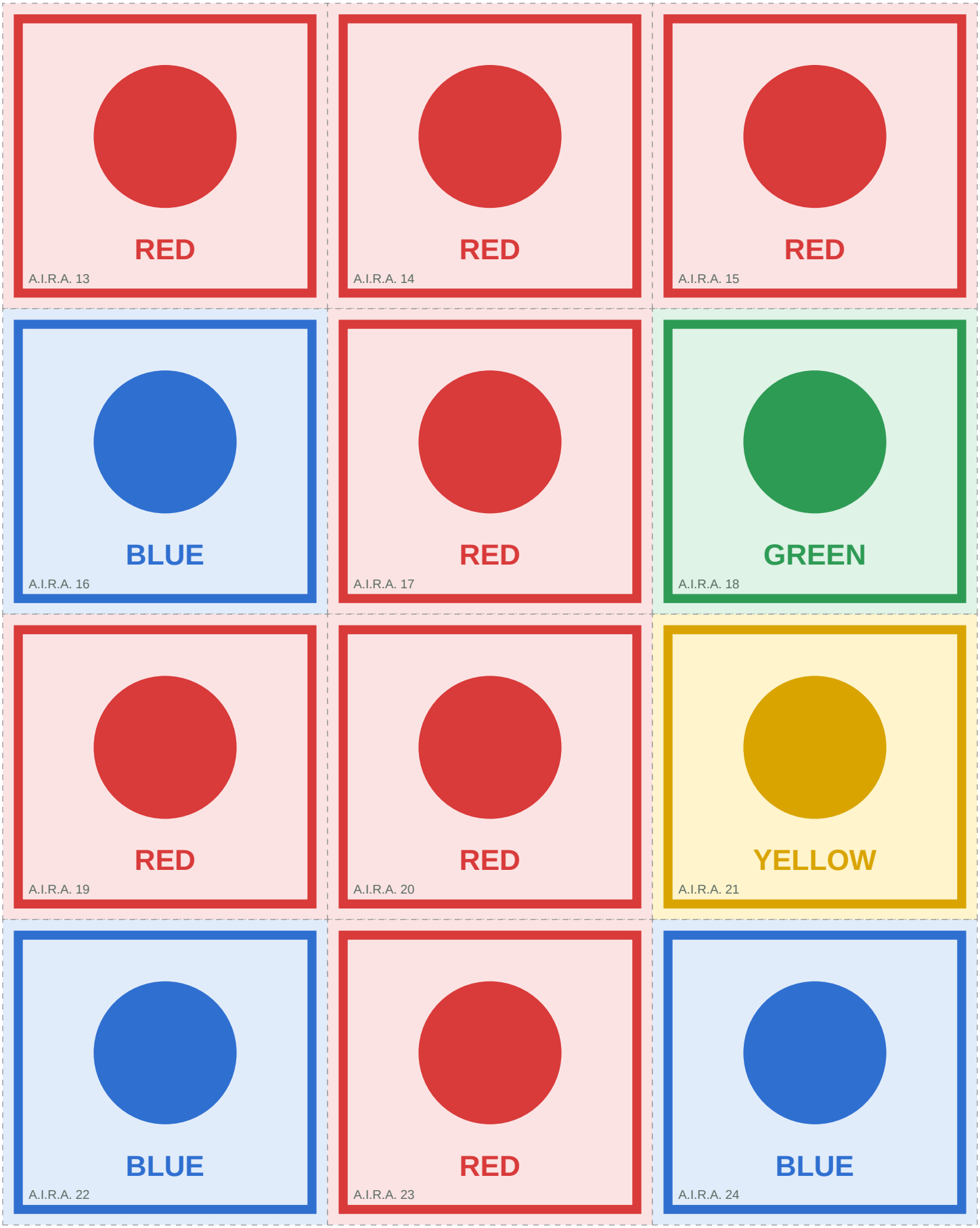
BIN B · EVERYTHING ELSE

Cut out the rule card and the two bin labels. Tape the labels to plates, cups or the floor, six feet apart.

CARD DECK, sheet 1 of 2. Cut on the dotted lines. 24 cards per squad. Color word printed on every card.



CARD DECK, sheet 2 of 2. Cut on the dotted lines. 24 cards per squad. Color word printed on every card.



Squad scorecard

one per squad

Squad: _____

Round	Role	Player	Seconds	Wrong bin	Red count (tally)	Correct count?
1	Machine					
1	Human					
2	Machine					
2	Human					

The pink card

What did the Machine do with the pink card?

What did the Human do with it?

Debrief

Where was the machine better?

Where was the machine worse?

Why do real workflows still need humans?

Check the red count against the deck: the correct red count is 10.

Role badges and clue word bank

cut out · one set per squad

INPUT 1

You see the card.
Call out ONE clue word.

INPUT 2

You see the card.
Call out ONE clue word.

HIDDEN 1

You do NOT see the card.
Hear both clues. Vote A or B.

HIDDEN 2

You do NOT see the card.
Hear both clues. Vote A or B.

OUTPUT

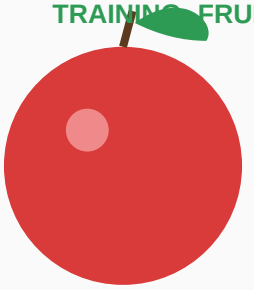
Hear both votes.
Announce the guess:
FRUIT or BALL.

CLUE WORD BANK (Input scouts, pick ONE word per card)

round	red	shiny	has a stem	has a leaf	curved
stripes	seams	bounces	yellow	orange	patches
sweet	hard	smooth	bumpy	white	green

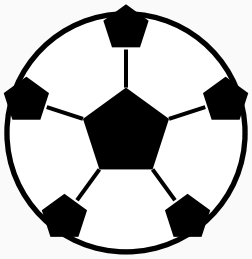
MYSTERY CARDS: 12 per squad. Only the Input scouts see the front. Cut on the dotted lines.

#1 TRAINING: FRUIT (A)



A.I.R.A. mystery card

#2 TRAINING: BALL (B)



A.I.R.A. mystery card

#3



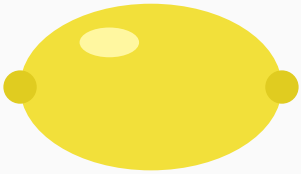
A.I.R.A. mystery card

#4



A.I.R.A. mystery card

#5

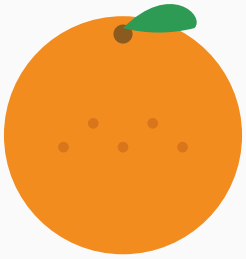


A.I.R.A. mystery card

#6

A.I.R.A. mystery card

#7



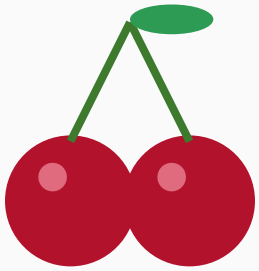
A.I.R.A. mystery card

#8



A.I.R.A. mystery card

#9



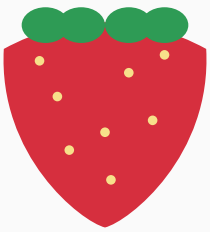
A.I.R.A. mystery card

#10



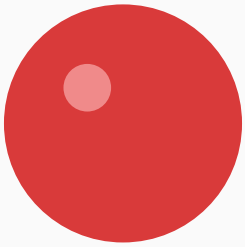
A.I.R.A. mystery card

#11



A.I.R.A. mystery card

#12



A.I.R.A. mystery card

Squad scorecard

one per squad

Squad: _____

Card #	Clue 1	Clue 2	Hidden 1	Hidden 2	Output guess	Right?
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

Reflect (card 12, the red ball)

What did the model guess, and was it right?

Which clue fooled the model? What clue was missing?

What would you add to the training data so it stops making this mistake?